Deterministic Optimization

Nonlinear Optimization Modeling – Approximation and Fitting

Andy Sun
Assistant Professor
Stewart School of Industrial and Systems Engineering

Function Fitting

Nonlinear Optimization Modeling

Learning Objectives for this lesson

- Discover how to use least squares to do data fitting with arbitrary polynomial functions

Data Fitting Problem

- Suppose we have m pairs of data points $(a_1, b_1), \dots, (a_m, b_m)$.
    - Think about an experiment that collects input and output of a device
    - We want to figure out a functional relation between a and b
- We want to find the "best" cubic polynomial fit b_i as a function of a_i
- A cubic polynomial can be written as
    - $p(a) = x_0 + x_1 a + x_2 a^2 + x_3 a^3$
- The error of using $b = p(a)$ to fit the data is:
    - $r(x) = [p(a_1) - b_1, \dots, p(a_n) - b_n]^\top$
    - $r(x)$ is a function of $x = (x_0, \dots, x_3)$ the coefficients of $p(a)$
- We want to find x such that norm of $r(x)$ is minimized

Error As a Linear Function of x

- Let us write the error term more explicitly:
- $r = \begin{bmatrix} p(a_1) - b_1 \\ \vdots \\ p(a_n) - b_n \end{bmatrix} = \begin{bmatrix} p(a_1) \\ \vdots \\ p(a_n) \end{bmatrix} - \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} x_0 + x_1 a_1 + x_2 a_1^2 + x_3 a_1^3 \\ \vdots \\ x_0 + x_1 a_n + x_2 a_n^2 + x_3 a_n^3 \end{bmatrix} - \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$

$$= \begin{bmatrix} 1 & a_1 & a_1^2 & a_1^3 \\ & & \vdots & \\ 1 & a_n & a_n^2 & a_n^3 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} - \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

$$= Ax - b$$

Use Least Squares to Find Best Data Fit

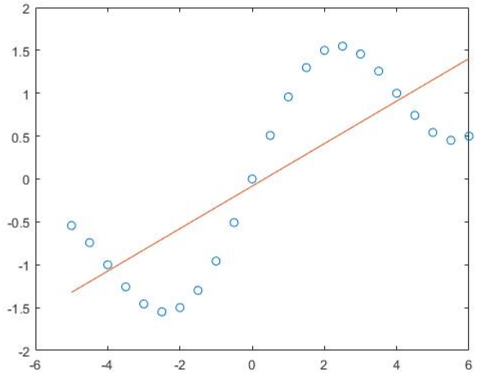
- Now we can apply Least Squares to minimize the l_2 norm of the error

$$\min_x ||Ax - b||_2^2$$

- Least squares can be solved very efficiently by numerical linear algebra method as solving linear equations.

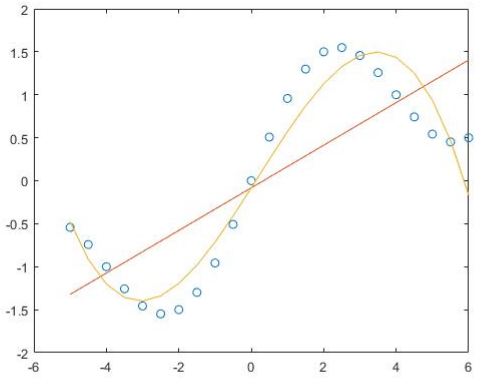
Example

- Example of fitting a polynomial to the following data:
    - $x = -5, -4.5, \dots, 5.5, 6$
    - $y = \sin\left(\frac{\pi x}{4}\right) + \frac{x}{4}$
- Data and deg-1 polynomial fitting:



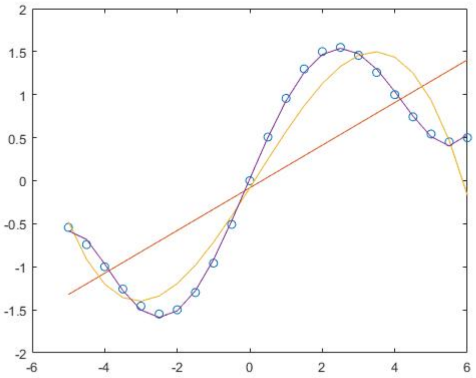
Example

- Example of fitting a polynomial to the following data:
  - $x = -5, -4.5, \dots, 5.5, 6$
  - $y = \sin\left(\frac{\pi x}{4}\right) + \frac{x}{4}$
- Data and deg-3 polynomial fitting:



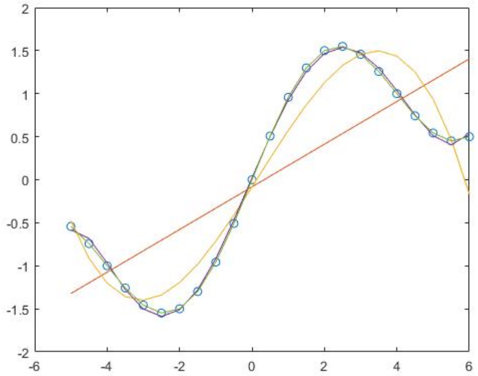
Example

- Example of fitting a polynomial to the following data:
  - $x = -5, -4.5, \dots, 5.5, 6$
  - $y = \sin\left(\frac{\pi x}{4}\right) + \frac{x}{4}$
- Data and deg-6 polynomial fitting:



Example

- Example of fitting a polynomial to the following data:
    - $x = -5, -4.5, \dots, 5.5, 6$
    - $y = \sin\left(\frac{\pi x}{4}\right) + \frac{x}{4}$
- Data and deg-12 polynomial fitting:



Summary

- We learned:
  - How to use least squares to solve function fitting problem
  - The key point is we can fit a highly nonlinear function to data as solving linear equations approximately through least squares